Week 11 BA

2024 Sep01 ET

1. [Source Q3; MCQ]
The objective function of the linear programming model using pair-wise judgments:

(a) Maximize the distance from the ideal point

(b) Minimize the distance from the ideal point

(c) Minimize the poorness of fit

(d) Both Maximize the distance from the ideal point & Minimize the poorness of fit

(e) Both Minimize the distance from the ideal point & Minimize the poorness of fit

[Source ans: c]

2. [Source Q10; MSQ]
What is the format of data needed for performing the conjoint analysis using the Statistical or Linear Regression Approach?

(a) Consumer Choice Data is Ratings

(b) Consumer Choice Data is Pairwise Comparison

(c) Value of the attributes are continuous

(d) Value of the product attributes are categorical

[Source ans: a, d]

2024 Dec22 ET

3. [Source Q37; MSQ]
What is the format of data needed for performing the conjoint analysis using the Statistical or Linear Regression Approach?

(a) Consumer Choice Data is Ratings

(b) Value of the attributes are continuous

(c) Consumer Choice Data is Pairwise Comparison

(d) Value of the product attributes are categorical

[Source ans: a, d]

4. [Source Q38; MCQ]
The objective function of the linear programming model using pair-wise judgments:

(a) Minimize the poorness of fit

(b) Maximize the distance from the ideal point

(c) Minimize the distance from the ideal point

(d) Both Minimize the poorness of fit & Minimize the distance from the ideal point

(e) Both Maximize the distance from the ideal point & Minimize the distance fromthe ideal point

[Source ans: a]

2025 Apr13 ET

5. [Source Q6; MSQ]
The part worth can be defined as (choose all that may be applicable)

(a) Level utilities

(b) The utility for that level of attribute

(c) Utility for separate parts of the products

(d) None of these

[Source ans: a, b, c]

2025 Aug31 ET

Comprehension for questions 6-9 (source Q35)

You are in the process of selecting an ideal Laptop when a lot of options are available. For example, the three brands that have been on top of your mind are Acer, HP, and Dell. Similarly, there are 2 RAM options available: 8 GB and 16 GB. Lastly, the processor is another important variable in deciding the ideal laptop. There are 3 processors available: i3, i5, and i7. Also, assume that Acer is the least preferred brand. Based on the inputs available, you build a regression model, and the coefficients are given below in Figure 1. Based on the above data, answer the given subquestions.

SUMMARY OUTPUT

Regression Statistics	
Multiple R	0.976
R Square	0.952
Adjusted R Square	0.932
Standard Error	0.667
Observations	18

ANOVA

	df	*SS*	*MS*	*F*	*Significance F*
Regression	5	105.1666667	21.03333333	47.325	1.73991E-07
Residual	12	5.333333333	0.444444444		
Total	17	110.5			

	Coefficients	*Standard Error*	*t Stat*	*P-value*	*Lower 95%*	*Upper 95%*	*Lower 95.0%*	*Upper 95.0%*
Intercept	1.333	0.385	3.464	0.005	0.495	2.172	0.495	2.172
HP	1.167	0.385	3.031	0.010	0.328	2.005	0.328	2.005
Dell	1.833	0.385	4.763	0.000	0.995	2.672	0.995	2.672
16 GB RAM	2.333	0.314	7.425	0.000	1.649	3.018	1.649	3.018
i5 processor	2.1667	0.385	12.557	0.000	3.995	5.672	3.995	5.672
i7 processor	4.8333	0.385	5.629	0.000	1.328	3.005	1.328	3.005

Figure 1: Regression Summary

6. [Source Q36; SA]
What is the part worth to customers if he/she upgrades from an i3 to an i7 processor? [Note: Enter your answer in decimal rounded to four decimal places. For example, if your answer is 1000.63545, then enter the answer as "1000.6355"]

[Source ans: 4.82 to 4.84]

7. [Source Q37; SA]
How much is the weightage provided by the customer for the Brand? [Note: Enter your answer in decimal rounded to four decimal places. For example, if your answer is 0.63545, then enter the answer as "63.55"]

[Source ans: 20.35 to 20.39]

8. [Source Q38; SA]
How much is the weightage provided by the customer for the RAM? [Note: Enter your answer in decimal rounded to four decimal places. For example, if your answer is 0.63545, then enter the answer as "63.55"]

[Source ans: 25.90 to 25.96]

9. [Source Q39; SA]
How much is the weightage provided by the customer for the processor? [Note: Enter your answer in decimal rounded to four decimal places. For example, if your answer is 0.63545, then enter the answer as "63.55"]

[Source ans: 53.65 to 53.75]

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2026 May06 ET

10. [Source Q21; MCQ]
The part worth can be defined as:

(a) Level utilities

(b) The utility for that level of attribute

(c) Utility for separate parts of the products

(d) All of these

[Source ans: d]

11. [Source Q22; MCQ]
In the pairwise comparison, which of the following is true:

(a) We want to minimize the violation with regard to wrong preferences

(b) Objective function is to maximize the goodness of fit

(c) Objective function is to minimize the poorness of fit

(d) Both We want to minimize the violation with regard to wrong preferences and Objective function is to minimize the poorness of fit

(e) None of these

[Source ans: d]

Comprehension for questions 12-15 (source Q23)

The following scenario is purely hypothetical You are in the process of selecting an ideal car from several available options. The decision depends on three key attributes: • Brand: Hyundai, Maruti Suzuki, and Volkswagen • Gearbox type: Manual and Automatic • Engine displacement (CC): 1000 CC, 1300 CC, and 1500 CC Your preferences are as follows: • Among brands, Hyundai is least preferred, followed by Maruti Suzuki, while Volkswagen is most preferred. • Automatic gearbox is preferred over Manual. • For engine capacity, 1500 CC > 1300 CC > 1000 CC in terms of performance and preference. Using these inputs, you build a regression model to quantify your preferences. The estimated coefficients of the model are provided in Figure 2.

SUMMARY OUTPUT								
Regression Statistics								
Multiple R	0.96							
R Square	0.94							
Adjusted R Square	0.92							
Standard Error	0.66							
Observations	18							
ANOVA								
	df	*SS*	*MS*	*F*	*Significance F*			
Regression	5	105.1666667	21.03333333	47.325	1.73991E-07			
Residual	12	5.333333333	0.444444444					
Total	17	110.5						
	Coefficients	*Standard Error*	*t Stat*	*P-value*	*Lower 95%*	*Upper 95%*	*Lower 95.0%*	*Upper 95.0%*
Intercept	1.33	0.38	3.46	0.00	0.49	2.17	0.49	2.17
Maruti	1.17	0.38	3.03	0.01	0.33	2.01	0.33	2.01
Volkswagen	1.83	0.38	4.76	0.00	0.99	2.67	0.99	2.67
Automatic	2.33	0.31	7.42	0.00	1.65	3.02	1.65	3.02
1300 CC	2.17	0.38	12.56	0.00	3.99	5.67	3.99	5.67
1500 CC	4.83	0.38	5.63	0.00	1.33	3.01	1.33	3.01

Figure 2: Regression Model Output

Based on the above data,answer the given sub-questions

12. [Source Q24; SA]
What is the part worth to customers if he/she upgrades from a 1000 CC to a 1500 CC engine? [Note: Enter your answer in decimal rounded to four decimal places. For example, if your answer is 1000.63545, then enter the answer as "1000.6355"]

[Source ans: 4.82 to 4.84]

13. [Source Q25; SA]
How much is the weightage provided by the customer for the Brand? (Note: Enter the answer in "%" rounded to two decimal places without the "%" symbol. For example, if the answer is "0.7354%", then enter it as "0.74")

[Source ans: 20.40 to 20.90]

14. [Source Q26; SA]
How much is the weightage provided by the customer for the Gearbox? (Note: Enter the answer in "%" rounded to two decimal places without the "%" symbol. For example, if the answer is "0.7354%", then enter it as "0.74")

[Source ans: 26.10 to 26.60]

15. [Source Q27; SA]
How much is the weightage provided by the customer for the Engine? (Note: Enter the answer in "%" rounded to two decimal places without the "%" symbol. For example, if the answer is "0.7354%", then enter it as "0.74")

[Source ans: 54.25 to 54.75]